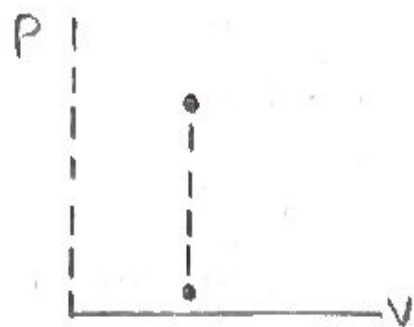


Q-1 Discuss following thermodynamics terms:-

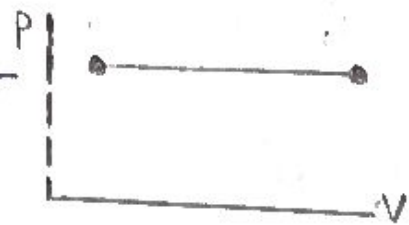
Ans- A-) Isochoric process:-

- volume remains constant $V = \text{constant}$
- No boundary work $W = \int P dV = 0$



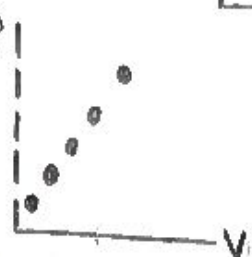
b-) Isobaric process:-

- pressure remains constant
- $W = P(V_2 - V_1)$



c-) Isothermal process:-

- Temp. constant
- $PV = \text{constant}$



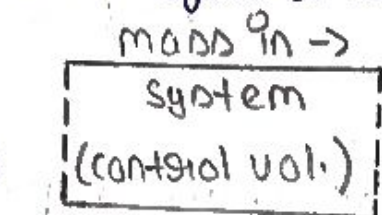
d-) quasi-static process:-

- process occurs very slowly
- system passes through sequence of equilibrium states



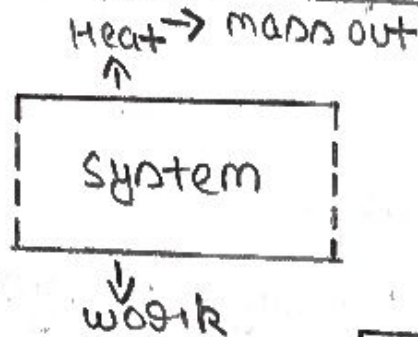
E-) open system:-

- mass and energy cross boundary
- Ex- Turbine, Pump



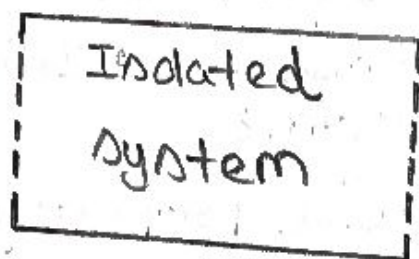
F-) closed system:-

- No mass crosses boundary
- Energy may cross



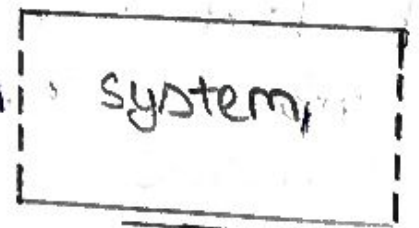
G-) Isolated system:-

- No mass and no energy transfer
- Ex- Ideal thermos flask



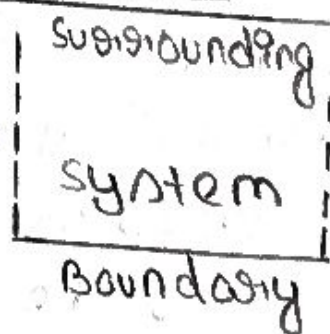
H-) system:-

- A specified quantity of matter or region in space chosen for study.



I-) Boundary:-

- Real or imaginary surface separating system and surroundings
- may be fixed or movable



J-) surroundings:-

- Everything outside the system boundary

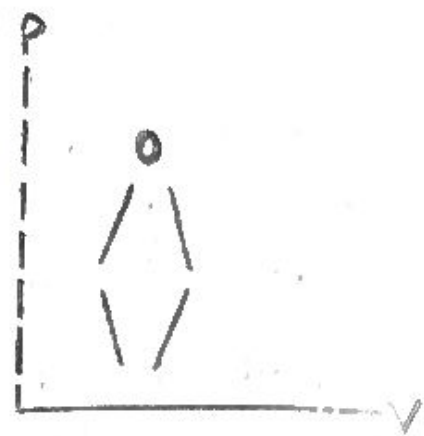
K) process:-

- A change of state of a system from one equilibrium state to another

L) Cyclic process:-

A process in which system returns to its initial state

- Net change in properties = 0
- $\oint du = 0$



M) Point function:-

A property that depends only on the state of the system, not on path

- Ex →
- pressure
 - Temp.
 - volume
 - Internal Energy

N) Path function:-

A quantity that depends on path followed during a process

- Ex →
- work
 - Heat

O) Extensive property:-

Depends on mass or size of system

- Ex →
- mass
 - Volume
 - Internal Energy

P) Intensive property:-

- Independent of mass

- Ex →
- Temp
 - pressure
 - Density

Q) PMM-1:-

A hypothetical machine that:-

- produces work without energy input
- violates the first law of thermodynamics

R) first law of Thermodynamics :- (for process)

$$Q = \Delta U + W$$

S-) first law for cyclic process :-
Since $\Delta U = 0$ in cycle :-

$$Q = W$$

T-) limitations of first law :-

- 1-) Does not indicate direction of heat flow.
- 2-) Does not explain irreversibility.
- 3-) Does not explain entropy

U-) microscopic point of view :-

- motion of atoms and molecules
- statistical mechanics
- Energy distribution

V-) macroscopic point of view :-

- Pressure
- Temp.
- Volume

N-) Role of Thermodynamics in quantum :-

- Thermodynamics applied at quantum scale
- 1-) Energy quantization
- 2-) quantum heat engines
- 3-) Nano-scale systems

X-) Zeroth law of thermodynamics :-

If • system A is thermal equilibrium with B

- B is thermal equilibrium with C

Then, A is thermal equilibrium with C

Y-) Thermodynamics Equilibrium :-

- 1-) Thermal equilibrium
- 2-) mechanical equilibrium
- 3-) chemical equilibrium

- 2-) Role of Thermodynamics in daily life:-
- Refrigerator and air conditioners
 - Heat engine
 - Cooking
 - Solar panels

Q-2)

Ans-) For an adiabatic process:-

$$P V^{\gamma} = C$$

$$P = \frac{C}{V^{\gamma}}$$

$$\Rightarrow W = \int_{V_1}^{V_2} \frac{C}{V^{\gamma}} dV$$

$$W = C \int_{V_1}^{V_2} V^{-\gamma} dV$$

$$\Rightarrow \int V^{-\gamma} dV = \frac{V^{1-\gamma}}{1-\gamma}$$

$$\Rightarrow W = \frac{C}{1-\gamma} (V_2^{1-\gamma} - V_1^{1-\gamma})$$

$$\Rightarrow C = P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$$

$$\Rightarrow W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

Q-3)

Ans-) $P V^n = C$

$$P = \frac{C}{V^n}$$

$$\Rightarrow W = \int_{V_1}^{V_2} \frac{C}{V^n} dV$$

$$\Rightarrow W = C \int_{V_1}^{V_2} V^{-n} dV$$

$$\Rightarrow W = \int V^{-n} dV = \frac{V^{1-n}}{1-n}$$

$$\Rightarrow W = \frac{C}{1-n} (V_2^{1-n} - V_1^{1-n})$$

$$\Rightarrow C = P_1 V_1^n = P_2 V_2^n$$

$$\Rightarrow W = \frac{P_1 V_1 - P_2 V_2}{n-1}$$

Q-4.)

Ans- a-) Derivation of steady flow Energy equation:-

Step-1 $Q - W = \frac{dE_{cv}}{dt} + \sum \dot{m}_{out} e_{out} - \sum \dot{m}_{in} e_{in}$

$$\Rightarrow Q - W = \sum \dot{m}_{out} e_{out} - \sum \dot{m}_{in} e_{in}$$

$$\Rightarrow Q - W = m \left[\left(u_2 + \frac{V_2^2}{2} + g z_2 \right) - \left(u_1 + \frac{V_1^2}{2} + g z_1 \right) \right]$$

$$\Rightarrow Q - W = m \left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g (z_2 - z_1) \right]$$

$$\Rightarrow Q - W = (h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g (z_2 - z_1)$$

b-) Assumptions for steady-flow operations:-

$$\Rightarrow \frac{dE_{cv}}{dt} = 0$$

$$\Rightarrow \dot{m}_{in} = \dot{m}_{out}$$

$$\Rightarrow g(z_2 - z_1) \approx 0$$

$$\Rightarrow Q \approx 0$$

$$\Rightarrow h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

Q-5)

Ans-)

Process	Q (kJ/hr)	W (kJ/hr)	ΔU (kJ/hr)
A-B	0	35	-35
B-C	-150	0	-150
C-D	400	200	200
D-E	400	-200	600
E-A	1,799,350	1,799,965	-615

Step-ii)

$$\Sigma W = 1,800,000 \text{ kJ/hr}$$

$$\Rightarrow \frac{1,800,000}{3600} \Rightarrow 500 \text{ kJ/sec}$$

Q-6

Ans-)

Process	Q (kJ/min)	W (kJ/min)	ΔU (kJ/min)
1-2	0	35	-35
2-3	-150	0	-150
3-4	400	200	200
4-5	400	-200	600
5-1	29,350	29,965	-615

$$ii-) \Sigma W = 30,000 \text{ kJ/min}$$

$$\text{convert to kJ/hour} = 30,000 \times 60$$

$$= 1,800,000 \text{ kJ/hour}$$

Q-7)

Ans-

Process	Q (kJ/sec)	W (kJ/sec)	ΔU (kJ/sec)
m-n	0	35	-35
n-o	-150	0	-150
o-p	400	200	200
p-q	400	-200	600
q-m	4350	4965	-615

Step-ii) Net Rate of work output

$$\Sigma W = \Sigma Q$$

$$\Rightarrow 5000 \text{ kJ/sec}$$

Q-8-)

Ans- At ice point $\rightarrow T_m = 32^\circ \text{F}$, $F = 24.441$

$$32 = \lambda (\ln 24.441) + \mu + 36.80$$

$$32 - 36.80 = \lambda (\ln 24.441) + \mu$$

$$-4.80 = \lambda (\ln 24.441) + \mu$$

At steam point $\rightarrow T_m = 212^\circ \text{F}$, $F = 50$

$$212 = \lambda (\ln 50) + \mu + 36.80$$

$$212 - 36.80 = \lambda (\ln 50) + \mu$$

$$175.20 = \lambda (\ln 50) + \mu$$

Solve for λ

$$175.20 - (-4.80) = (3.912 - 3.196)\lambda$$

$$180 = 0.716\lambda$$

$$\lambda \approx 251.4$$

substitute into first eq, $-4.80 = 3.196(251.4) + \mu$
 $-4.80 = 803.5 + \mu$
 $\mu = -808.3$

$$F = 36.80$$

$$(36.80) \approx 3.605$$

$$T_m = 251.4(3.605) - 808.3 + 36.80$$

$$T_m = 906.3 - 808.3 + 36.80$$

$$T_m \approx 134.8^\circ \text{F}$$

$$\Rightarrow T_m \approx 135^\circ \text{F}$$

8-9.)

Ans- $T_m = \lambda F + \mu + T_c$

At ice point, $T_m = 250 = \lambda(25) + \mu + 50$

$$250 - 50 = 25\lambda + \mu$$

$$200 = 25\lambda + \mu \quad (1)$$

At steam point,

$$T_m = 500 = \lambda(50) + \mu + 50$$

$$500 - 50 = 50\lambda + \mu$$

$$450 = 50\lambda + \mu \quad (2)$$

Now, subtract (1) from (2)

$$450 - 200 = (50 - 25)\lambda$$

$$250 = 25\lambda$$

$$\lambda = 10$$

substitute λ into (1):-

$$200 = 25(10) + \mu$$

$$200 = 250 + \mu$$

$$\mu = -50$$

$$T_m = \lambda F + \mu + T_c$$

$$T_m = 10(36.80) + (-50) + 50$$

$$T_H = 368^\circ\text{S}$$

$$\Rightarrow T_H \approx 368^\circ\text{S}$$

Q-10)

Ans-

Ice point, $0 = n \ln(51) + \lambda$

$$\lambda = -n \ln(51)$$

Steam point, $100 = n \ln(100) + \lambda$

Substitute $\lambda = -n \ln(51)$ $100 = n \ln(100) - n \ln(51)$

$$100 = n \ln\left(\frac{100}{51}\right)$$

solve for n ; $\ln\left(\frac{100}{51}\right) = \ln(1.96078) \approx 0.673$

$$n = \frac{100}{0.673} \approx 148.6$$

solve for λ , $\lambda = -n \ln(51)$

$$\ln(51) \approx 3.932$$

$$\lambda = -148.6 \times 3.932 \approx -583.5$$

Now, $T = n \ln(138) + \lambda$

$$\ln(138) \approx 4.927$$

$$T \approx 732.1 - 583.5 \approx 148.6^\circ\text{C}$$

$$T \approx 149^\circ\text{C}$$

Q-11

Ans- Given, $P_1 = 3 \times 10^5 \text{ Pa}$

$$V_1 = 3 \text{ m}^3$$

$$P_2 = 1.2 \times 10^5 \text{ Pa}$$

$$n = 2$$

$$W = \frac{C}{1-n} (V_2^{1-n} - V_1^{1-n}), \quad n \neq 1$$

$$C = P_1 V_1^n = 3 \times 10^5 \times (3)^2$$

$$C = 3 \times 10^5 \times 9 = 2.7 \times 10^6 \text{ Pa}$$

$$V_2^2 = \frac{C}{P_2} = \frac{2.7 \times 10^6}{1.2 \times 10^5}$$

$$V_2^2 = 22.5$$

$$V_2 = \sqrt{22.5} \approx 4.743 \text{ m}^3$$

calculate work, $\bar{W} = \frac{C}{-1} (V_2^{-1} - V_1^{-1}) = -C \left(\frac{1}{V_2} - \frac{1}{V_1} \right)$

$$\Rightarrow W = -2.7 \times 10^6 \left(\frac{1}{4.743} - \frac{1}{3} \right)$$

$$\Rightarrow \frac{1}{4.743} \approx 0.2108, \frac{1}{3} = 0.3333$$

$$\Rightarrow \frac{1}{V_2} - \frac{1}{V_1} = 0.2108 - 0.3333 = -0.1225$$

$$\Rightarrow W = -2.7 \times 10^6 \times (-0.1225)$$

$$\Rightarrow W \approx 330,750 \text{ J} \approx 0.331 \text{ MJ}$$

$$\boxed{W \approx 0.331 \text{ MJ}}$$

0-12)

Ans-) Given, $P_1 = 2.5 \times 10^5 \text{ Pa}$

$$V_1 = 36 \text{ m}^3, V_2 = 100 \text{ m}^3$$

$$n = 2$$

$$W = -C \left(\frac{1}{V_2} - \frac{1}{V_1} \right)$$

$$\Rightarrow C = P_1 V_1^2 = 2.5 \times 10^5 \times (36)^2$$

$$36^2 = 1296$$

$$C = 3.24 \times 10^9 \text{ Pa}$$

Calculate $\frac{1}{V_2} - \frac{1}{V_1}$

$$\Rightarrow \frac{1}{100} - \frac{1}{36} \approx 0.01 - 0.02778$$

$$\Rightarrow -0.01778$$

$$W = -(3.24 \times 10^9)(-0.01778)$$

$$W \approx 5.76 \times 10^7 \text{ J} = 57.6 \text{ MJ}$$

$$W \approx 57.6 \text{ MJ}$$

Q-13)

Ans- $PV = 25 \Rightarrow V = \frac{25}{P}$

$$V = \frac{25}{P} \Rightarrow dV = -\frac{25}{P^2} dP$$

$$W = \int_{P_1}^{P_2} P dV = \int_5^{10} P \left(-\frac{25}{P^2} dP \right)$$

$$W = -25 [\ln P]_5^{10} = -25(\ln 10 - \ln 5)$$

$$W = -25 \ln 2$$

$$W = -25 \times 0.693 \approx -17.325 \text{ bar}$$

$$W \approx -1733 \text{ kJ}$$

Q-14)

Ans- $P_1 = 9.25 \times 10^5 \text{ Pa} = 9.25 \text{ bar}$

$$P_2 = 3.40285 \times 10^5 \text{ Pa}$$

$$W = -15 \int_{P_1}^{P_2} \frac{1}{P} dP$$

$$W = -15 [\ln P]_{P_1}^{P_2} = -15 (\ln P_2 - \ln P_1) = -15 \ln \left(\frac{P_2}{P_1} \right)$$

$$\frac{P_2}{P_1} = \frac{3.40285}{9.25} \approx 0.3678$$

$$\ln = (0.3678) \approx -1.000$$

$$W = -15 \times (-1.000) \approx 15 \text{ kJ}$$

$$W = 15 \times 100 \text{ kJ} = 1500 \text{ kJ}$$

$$W \approx 1500 \text{ kJ}$$

Q15)

Ans-) Sum the given heat transfers,

$$1-) \text{ group 1} \rightarrow 100 + 9 + 9 + 8 + 8 + 7 + 7 \Rightarrow 148 \text{ kJ}$$

$$2-) \text{ group 2} \rightarrow 100 - 9 - 9 - 8 - 8 - 7 - 7 \Rightarrow 42 \text{ kJ}$$

$$3-) \text{ group 3} \rightarrow 111 + 222 + 333 + 0 - 333 - 222 - 111 \Rightarrow 0 \text{ kJ}$$

$$4-) \text{ group 4} \rightarrow 6 + 6 + 5 + 4 + 3 + 2 + 2 \Rightarrow 28 \text{ kJ}$$

$$5-) \text{ group 5} \rightarrow 101 + 202 + 303 + 0 - 101 - 202 - 303 \Rightarrow 0 \text{ kJ}$$

$$6-) \text{ group 6} \rightarrow -6 - 6 - 5 - 4 - 3 - 2 - 2 \Rightarrow -28 \text{ kJ}$$

$$\text{Now, } Q_{\text{net}} \Rightarrow 148 + 42 + 28 - 28$$

$$\Rightarrow 190 \text{ kJ}$$

$$W_{\text{net}} = Q_{\text{net}} \Rightarrow 190 \text{ kJ}$$

Net work done per cycle, $W_{\text{net}} = 190 \text{ kJ}$

Net change in internal energy, $\Delta U_{\text{cycle}} = 0 \text{ kJ}$

Q-16)

Ans-) sum all work values :-

$$\text{group-1} \rightarrow 5+7+8+10+12+14+16+18+19+20 \Rightarrow 129 \text{ kJ}$$

$$\text{group-2} \rightarrow 4-5-7-8-10-12-14-16-18 \Rightarrow -86 \text{ kJ}$$

$$\text{group-3} \rightarrow 3-4+6+7+9+11+13+15+17-19 \Rightarrow 58 \text{ kJ}$$

$$\text{group-4} \rightarrow 2-2-3-6-7-9-11-13-15-17 \Rightarrow -81 \text{ kJ}$$

$$W_{\text{net}} \Rightarrow 129 - 86 + 58 - 81 \\ \Rightarrow 20 \text{ kJ}$$

$$Q_{\text{net}} \Rightarrow 20 \text{ kJ}$$

$$\Delta U_{\text{cycle}} \Rightarrow 0 \text{ kJ}$$

Q-17)

Ans-) sum of heat transfers

$$Q (\text{kJ/min}) = 5+10+15+15+20+25+30+35+40+45$$

$$Q_{\text{net}} = 240 \text{ kJ/min}$$

convert to kW

$$1 \text{ kJ/min} = \frac{1000 \text{ J}}{60 \text{ s}} \approx 16.6667 \text{ W}$$

$$W_{\text{net}} = 240 \times 16.6667 \approx 4000 \text{ W} = 4 \text{ kW}$$

$$W_{\text{net}} = 4 \text{ kW}$$

$$\Delta U_{\text{cycle}} = 0 \text{ kJ}$$